

Kubernetes and Cloud Native Associate (KCNA) Training - Labs

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Each folder contains its detailed guide, mock data, Kubernetes manifests and a verification script. Start with Lab 01 and the Learner Guide for workstation setup. The guide in each folder lists its own prerequisites and expected evidence.

Use the [Killercoda Kubernetes Playground](#) as the recommended browser-based cluster. Clone the public learner repository in its terminal, enter courseware/labs, then open the assigned lab folder. For advanced exercises that require extra cluster components or a longer-lived session, follow the lab-specific local kind fallback in the Learner Guide.

Day 1 - Cloud Native Foundations & Kubernetes Core Concepts

- [Lab 01 - Cluster Reconnaissance with kubect](#)
- [Lab 02 - Authoring Your First Pod Manifest](#)
- [Lab 03 - Commands, Arguments and Environment](#)
- [Lab 04 - Multi-Container Pod Patterns: Sidecar, Adapter, Ambassador](#)
- [Lab 05 - Container Images, OCI and the Runtime Interface](#)

Day 2 - Workloads, Scheduling & Container Orchestration

- [Lab 06 - Labels, Selectors and ReplicaSets](#)
- [Lab 07 - Deployments, Rollout Strategy and Rollback](#)
- [Lab 08 - DaemonSets, Jobs and CronJobs](#)
- [Lab 09 - Scheduling: nodeSelector, Affinity, Taints and Tolerations](#)
- [Lab 10 - Resource Requests, Limits, QoS and Autoscaling](#)

Day 3 - Services, Networking & Cluster Security

- [Lab 11 - Services, Endpoints and Cluster DNS](#)
- [Lab 12 - Service Types: ClusterIP, NodePort and LoadBalancer](#)
- [Lab 13 - Ingress Resources and HTTP Routing](#)
- [Lab 14 - Namespaces, ServiceAccounts and RBAC](#)
- [Lab 15 - Secrets, ConfigMaps and Safe Injection](#)
- [Lab 16 - NetworkPolicy and Default-Deny Isolation](#)

Day 4 - Storage, Cluster Architecture & Application Delivery

- [Lab 17 - Volumes: emptyDir, hostPath and the Container Filesystem](#)
- [Lab 18 - PersistentVolumes, Claims and StorageClasses](#)
- [Lab 19 - StatefulSets, Headless Services and Stable Identity](#)
- [Lab 20 - Control Plane Anatomy and etcd Backup/Restore](#)

- [Lab 21 - Packaging and Delivery: Helm, Kustomize and GitOps](#)

Day 5 - Observability, Troubleshooting & Assessment

- [Lab 22 - Probes, Health and Self-Healing](#)
- [Lab 23 - Events, Logs, Field Selectors and the Metrics Server](#)
- [Lab 24 - Metrics, Prometheus Exposition and the Observability Pipeline](#)
- [Lab 25 - Troubleshooting Triage: Application, Node and Control Plane](#)

Some advanced exercises use additional components. Lab 13 supplies a pinned Ingress controller path, Lab 16 supplies a policy-enforcing cluster path, and Lab 21 explains the tools needed to render Helm and Kustomize packages. Follow those lab-specific instructions before running the related checks.